

Deformable Object Simulation and Parameter Identification using Depth Observations

Antonio Škara, Bruno Maric, Frano Petric and Matko Orsag

Abstract—Physics-based models for bimanual manipulation of deformable materials must connect camera observations, tool motion, and material response. We present TaichiDough, a real-to-simulation pipeline for dough observed by one calibrated RGB-D camera while two spatulas follow motion-captured trajectories. Metric camera-scene calibration and floor-aware reconstruction initialize a mass-preserving three-dimensional Moving Least Squares Material Point Method model. Registered tool geometries are replayed as signed-distance fields, and material variables are fitted using a differentiable partial-view depth objective. On one recorded sequence, the saved history of a preliminary five-parameter run reduced its training objective by 61.8% over 15 accepted updates and reached a late-stage plateau. A separate fixed-configuration run improved visible-geometry scores relative to frozen initial geometry during the same recorded motion. Because the experiment uses one view, one trajectory, and no force measurements, the estimates are interpreted as model- and experiment-dependent effective parameters rather than unique dough constants. The results demonstrate an end-to-end observation-conditioned fitting pipeline and identify the measurements and validation needed for future robot-controlled data collection and synthetic-data generation.

I. INTRODUCTION

Robotic manipulation of dough and other elasto-plastic materials is difficult because every action changes both the visible geometry and the latent mechanical state. Tool contact can cause large deformation, rate-dependent response, adhesion, and irreversible shape change, while a single RGB-D camera observes only the exposed surface. These effects make broad physical data collection expensive and allow errors in reconstruction, contact, and constitutive response to produce similar image-space discrepancies [1]–[5]. A calibrated simulator can support controlled data generation and model-based action selection, but its predictions must first be tied to real measurements.

This paper studies observation-conditioned fitting of a specified three-dimensional simulator. TaichiDough reconstructs the initially visible dough surface into a volumetric particle model, replays two motion-captured rigid spatulas in MLS-MPM, renders the resulting particles in the calibrated camera, and adjusts material-model variables to reduce disagreement with recorded depth. Figure 1 shows the reconstructed object and registered tool geometries. The present experiment uses two spatulas, while the rigid signed-distance-field representation can accommodate other registered rigid tools; deformable end effectors would require an additional deformation model.

All authors are with the LARICS Laboratory for Robotics and Intelligent Control Systems, University of Zagreb, Faculty of Electrical Engineering and Computing, Zagreb 10000, Croatia. E-mail: {antonio.skara, bruno.maric, frano.petric, matko.orsag}@fer.hr.

The fitted values are effective parameters of the declared reconstruction, contact law, constitutive model, discretization, and recorded motion rather than universal dough constants [6], [7]. Robot execution and learned shaping policies are not evaluated here. The current emphasis is the calibrated sensing, replay, simulation, and fitting procedure that such downstream applications require.

The contributions are:

- a metric single-view reconstruction and tool-replay pipeline combining calibrated RGB-D observations, motion-captured dual-tool motion, and mass-preserving particle initialization;
- a differentiable partial-view objective and bounded parameterization for fitting elastic, viscous, and principal-stretch variables in a three-dimensional MLS-MPM rollout; and
- an experimental analysis on one recorded sequence that quantifies objective reduction and visible-geometry improvement while documenting the resulting observation, calibration, and identifiability limitations.

II. RELATED WORK

Deformable object manipulation. Deformable object manipulation includes linear, sheet-like, and volumetric objects such as ropes, cloth, food materials, and elasto-plastic media [1]–[3]. Their state is only partially observable and evolves through contact, deformation history, and material response. Existing approaches include model-based planning [8], adaptive visual servoing that estimates local motion-deformation relationships online [9], and learned predictive models or policies trained from demonstrations and interaction data [3]. G-DOOM combines learned graph dynamics with model-predictive control for rope and cloth manipulation [10]. The present work instead concerns physics-based real-to-simulation fitting for a volumetric material.

Deformable simulation and differentiable physics. Simulation approaches include mass-spring systems, finite-element methods, position-based dynamics, smoothed particle hydrodynamics, and the Material Point Method (MPM) [2], [3], [11]. MPM is well suited to large volumetric deformation because material points retain state while computation is transferred through a background grid, avoiding a permanently deforming mesh [11]. MLS-MPM and APIC provide the transfer formulations used here [12], [13]. ChainQueen and DiffTaichi demonstrate gradient computation through physical simulation [14], [15], while PlasticineLab applies differentiable simulation to elasto-plastic manipulation [16].

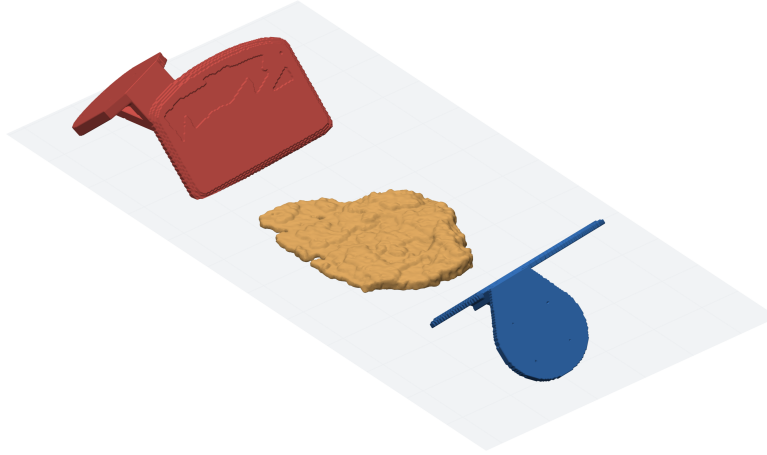


Fig. 1. Simulation scene with the reconstructed dough and registered watertight signed-distance-field collision geometries of the two spatulas.

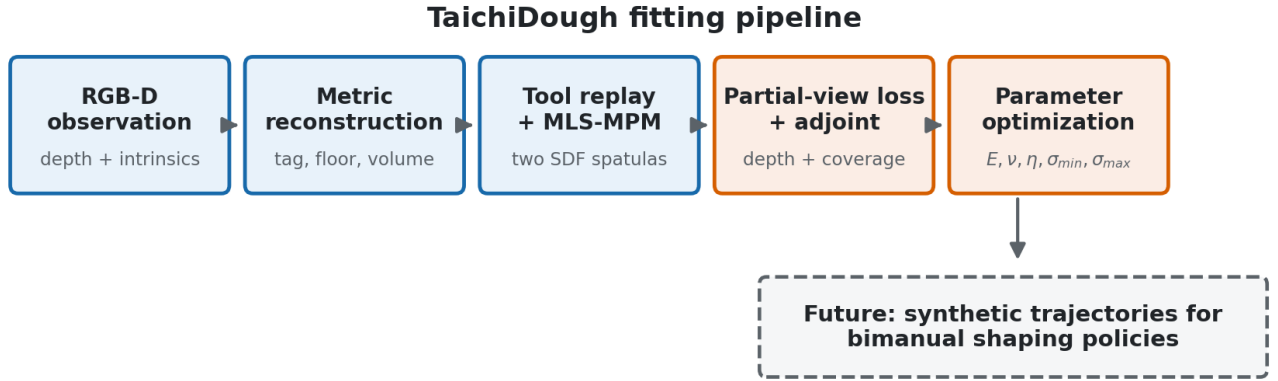


Fig. 2. TaichiDough pipeline. A calibrated RGB-D camera and recorded poses of two spatulas provide the initial geometry, observations, and boundary motion. An MLS-MPM rollout is projected into the camera and compared with measured depth. Robot execution and synthetic-data learning are downstream applications rather than experiments reported here.

TaichiDough uses these ideas to fit a prescribed simulator against partial depth observations.

Visual real-to-simulation fitting. Visual fitting methods differ primarily in observation and geometry assumptions. DiffCloud fits deformable simulations to point clouds, whereas Differentiable Depth uses a single depth camera with known reference geometry [17], [18]. PAC-NeRF jointly estimates geometry and physical parameters from multi-view video [19], and PhysTwin reconstructs spring-mass digital twins from interaction videos [20]. DPSI fits elasto-plastic material and contact parameters from real interactions and incomplete point clouds [21], while EMPM combines multi-view RGB-D reconstruction with online MPM refinement [22]. TaichiDough examines a more restricted measurement setting: one calibrated depth view, a floor-filled initial volume, and measured dual-tool motion. This reduces sensing requirements but leaves hidden geometry, contact, and material effects mutually ambiguous.

Robotic shaping and dough mechanics. RoboCraft and RoboCook learn predictive dynamics for goal-directed elasto-

plastic shaping with diverse tools [23], [24], while DoughNet predicts geometric and topological changes during dough manipulation [25]. Food-engineering studies instead estimate dough rheology from controlled mechanical measurements; Fabbri and Cevoli invert a finite-element model using extrusion measurements [26]. Dough response also depends on formulation, preparation, loading history, and adhesion [4], [5]. In contrast, the present experiment fits simulator parameters from visual motion without force measurements. Its values characterize the selected model and recorded sequence, not the complete rheology of the dough.

III. METHOD

TaichiDough processes a synchronized RGB-D and motion-capture recording in four stages: metric scene reconstruction, rigid-tool replay, MLS-MPM simulation, and partial-view parameter fitting. Figure 2 summarizes this sequence. The experimental setup is described first, followed by the simulator and optimization objective.



Fig. 3. Manual data collection with two 3D-printed spatulas. OptiTrack markers are mounted on the spatulas, while the RGB-D camera observes the dough and tools.

A. Experimental Setup and Data Collection

A human operator shaped one dough specimen using two 3D-printed spatulas, as shown in Fig. 3. An Intel RealSense D435 recorded aligned color and depth, and an OptiTrack motion-capture system recorded the rigid marker poses attached to both tools. Camera and motion-capture samples were associated by their recorded timestamps. A fixed rigid transform between each marker frame and its registered collision mesh was determined offline and then held constant throughout a replay.

Let z_t denote the simulated particle state, u_t the recorded poses and pose-derived velocities of the two spatulas, θ the fitted material variables, and ψ the fixed reconstruction, contact, camera, and numerical settings. For grid spacing Δx , time step Δt , and calibrated camera $\mathcal{C}$, the trajectory-conditioned inverse problem is

$$\begin{aligned} z_{t+1} &= \mathcal{S}_{\Delta x, \Delta t}(z_t, u_t; \theta, \psi), \\ y_t &= \mathcal{H}_{\mathcal{C}}(z_t) + \delta_t + \epsilon_t, \end{aligned} \quad (1)$$

where $\mathcal{H}_{\mathcal{C}}$ projects particles into the calibrated depth camera, while δ_t and ϵ_t denote model discrepancy and measurement noise. The fitted values are consequently effective parameters of this experiment and simulator rather than direct rheological measurements.

B. Metric Reconstruction and Tool Replay

Camera intrinsics and a rigid camera-to-motion-capture transform are obtained from repeated AprilTag/motion-capture correspondences. A second fixed rigid transform aligns the measured table plane with the horizontal scene plane. Every point cloud, tool pose, and camera pose is expressed in this metric table-aligned frame; no scale, translation, or post-hoc rigid registration is fitted during evaluation.

The first observed dough surface is voxelized, and each occupied vertical column is filled down to the calibrated table. For reconstructed volume V , fixed density ρ , and N material

points,

$$M = \rho V, \quad V_p = V/N, \quad m_p = \rho V/N. \quad (2)$$

The preliminary history run uses $\rho = 1200 \text{ kg m}^{-3}$, while the separate run visualized in Fig. 4 uses $\rho = 2196.219 \text{ kg m}^{-3}$; density is fixed within each configuration. This construction preserves the represented volume and derived mass when the particle count changes.

Tool positions are linearly interpolated, while quaternion samples are sign-aligned, interpolated, and normalized. Each spatula is represented by a signed-distance field generated from its registered collision mesh. Its velocity at contact point x is

$$v_{\text{tool}}(x) = v_{\text{lin}} + \omega \times (x - x_c), \quad (3)$$

where x_c is the tool center. The implemented unilateral contact update removes inward normal relative velocity and limits tangential correction using the configured Coulomb-friction coefficient; separating motion is unchanged, following the common sticking/sliding/separating treatment used in MPM contact [27].

C. Differentiable MLS-MPM

The simulator implements three-dimensional MLS-MPM with APIC transfer [12], [13] in Taichi [28]. Each particle stores position x_p , velocity v_p , deformation gradient F_p , and affine velocity matrix C_p . A trial deformation is first computed as

$$F_p^* = (I + \Delta t C_p^n) F_p^n = U \Sigma V^T, \quad (4)$$

where U and V contain the left and right singular vectors and $\Sigma = \text{diag}(\sigma_1, \sigma_2, \sigma_3)$ contains the principal stretches. In the preliminary five-parameter configuration, each stretch is projected according to

$$\tilde{\sigma}_i = \text{clip}(\sigma_i, \sigma_{\min}, \sigma_{\max}), \quad F_p^{n+1} = U \text{diag}(\tilde{\sigma}_i) V^T. \quad (5)$$

Thus $\sigma_{\min}$ and $\sigma_{\max}$ are dimensionless stretch limits, not yield stresses.

Young's modulus E and Poisson ratio ν define the Lamé parameters

$$\mu = \frac{E}{2(1 + \nu)}, \quad \lambda = \frac{E\nu}{(1 + \nu)(1 - 2\nu)}. \quad (6)$$

With polar factor R , $J = \det(F_p)$, and viscosity η , the implemented Kirchhoff stress contribution is

$$\tau = 2\mu(F_p - R)F_p^T + \lambda J(J - 1)I + \eta(C_p + C_p^T). \quad (7)$$

The final term is an additive rate stress rather than a multi-time-scale rheological model. For quadratic B-spline weight w_{pi} , particle-to-node offset d_{pi} , grid spacing Δx , particle volume V_p , and particle mass m_p , particle-to-grid transfer accumulates

$$m_i += w_{pi} m_p, \quad (8)$$

$$m_i v_i += w_{pi} [m_p v_p + (-4\Delta t V_p \Delta x^{-2} \tau + m_p C_p) d_{pi}]. \quad (9)$$

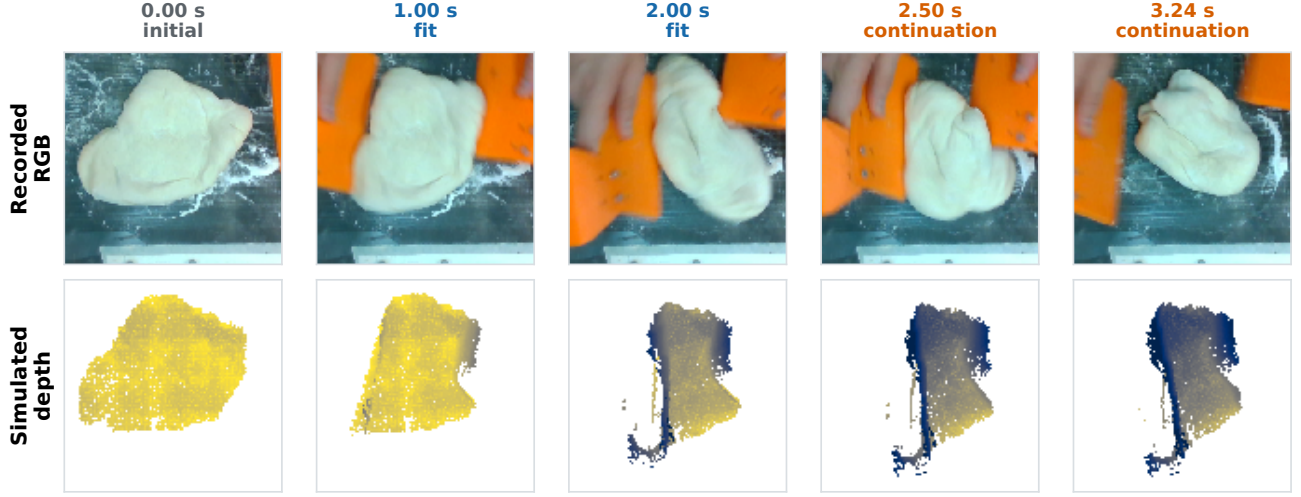


Fig. 4. Recorded RGB views and simulated dough-only depth from a separate strictly evaluated Episode 18 run at matched timestamps. The 0.00 s frame is the initial state; 1.00 and 2.00 s belong to its fitting interval, while 2.50 and 3.24 s are later frames of the same replay. The simulator uses the calibrated camera and display crop but does not render spatula occlusion. The later frames are temporal continuation, not an independent test trajectory.

The grid-to-particle update reconstructs velocity and affine motion as

$$v_p^{n+1} = \sum_i w_{pi} v_i, \quad C_p^{n+1} = 4\Delta x^{-2} \sum_i w_{pi} v_i d_{pi}^T. \quad (10)$$

D. Partial-View Objective and Optimization

At selected frames, a differentiable camera-space particle renderer uses compact splats and soft visibility to produce depth, coverage, and visible particle coordinates. The trajectory objective averages

$$\mathcal{L} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} (0.5\ell_d^t + 0.25\ell_c^t + 0.25\ell_q^t), \quad (11)$$

where ℓ_d applies a coverage-blended robust depth-change residual on valid fixed-support pixels and uses a normalized missing-depth residual of 4.0 where predicted support is absent, ℓ_c penalizes missing positive observed foreground, and ℓ_q is the directed distance from observed points to visible predicted particles. Measured depth residuals and point distances use a 0.01 m scale before the robust penalty. Invalid or unobserved camera pixels are masked out rather than interpreted as measured free space.

Checkpointed reverse-mode differentiation computes gradients of the trajectory loss while bounding device memory. Physical variables are represented by bounded, scaled optimizer coordinates; Young’s modulus uses a logarithmic coordinate because of its broad admissible range. A bound-constrained optimizer with backtracking rejects invalid or non-decreasing trial steps. The preliminary run fits E , ν , η , $\sigma_{\min}$, and $\sigma_{\max}$ jointly. Its checkpoint recomputation permits small state and contact-summary mismatches, so the resulting gradients are treated as approximate optimization evidence rather than qualified exact full-trajectory derivatives.

IV. RESULTS AND DISCUSSION

The stored Episode 18 diagnostics use distinct fixed configurations, so their losses and parameter values are not pooled across runs. Figure 5 reports the preliminary five-parameter history, whereas Fig. 4 shows outputs from a separate strictly evaluated run.

A. Preliminary Five-Parameter Fitting

The saved history contains the initial state and 15 accepted updates. The dimensionless partial-view training objective decreased from 0.42455 to 0.16229, a reduction of 61.8%. Most of the reduction occurred by update 7; updates 10–15 changed the objective from 0.16322 to 0.16229, a further 0.57%, indicating a late-stage plateau in the saved history. The lowest-loss saved state is

$$E = 8.80 \text{ kPa}, \quad \nu = 0.487, \quad \eta = 30.4 \text{ Pa s}, \quad (12)$$

$$\sigma_{\min} = 0.8626, \quad \sigma_{\max} = 1.0671.$$

The variables move together during optimization: viscosity increases, both stretch limits shift downward with little change in interval width, and ν approaches the upper part of its allowed range. This coordinated change is consistent with compensation among constitutive effects under one trajectory. The values in (12) are therefore a preliminary effective-parameter result, not final material calibration.

B. Visible-Geometry Evaluation

A separate five-parameter run with 20 accepted updates supplies Fig. 4. Its selected state reached the $E = 300$ kPa upper bound and the $\eta = 0$ lower bound. A strict full-resolution evaluator equally averages normalized depth-change error, one minus mask intersection-over-union, directed observed-to-simulation distance, and a missing-coverage penalty. Its mean score, for which lower is better, was 0.2821 compared with 0.3416 for frozen initial geometry on frames 1–60. On frames 61–97 of the same replay, the corresponding

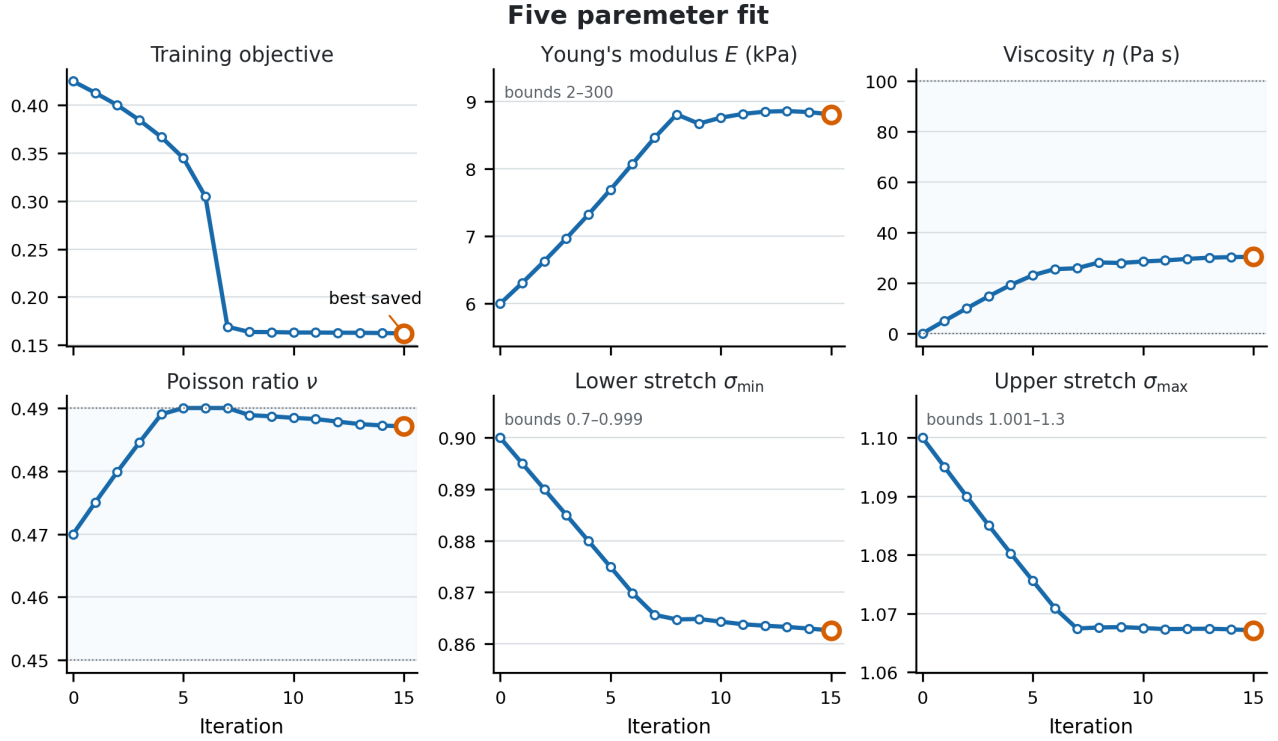


Fig. 5. Accepted-state objective and parameter history for the preliminary five-parameter run. The orange ring marks the lowest-loss saved state; all 16 retained states are shown without smoothing.

scores were 0.3939 and 0.8265; all 60 fitting frames and 37 continuation frames were valid. The run therefore improves visible-geometry prediction relative to a static-object baseline during the recorded motion, but the parameter bounds prevent interpreting it as calibrated physical recovery.

Figure 4 shows qualitative agreement in gross visible geometry through 1.00 s. By 2.00 s, the simulated visible geometry differs markedly from the recording, and little subsequent deformation is apparent in the simulation. This comparison is descriptive because the rows show different modalities and simulated depth does not include tool occlusion.

C. Limitations

A single depth view leaves hidden volume and occluded contact unmeasured, and the evaluator does not render the spatulas into simulated depth. Both reported runs use the fixed table-aligned calibration described in Sec. III; the estimates nevertheless remain conditional on residual camera, floor, and tool-registration uncertainty, reconstructed volume and density, contact law and friction, numerical resolution, and loss smoothing. Near-incompressible ν , elasticity, viscosity, and stretch clipping can also compensate for one another under a single trajectory.

No tool force or torque measurements were recorded. The optimizer therefore distinguishes contact and constitutive effects only through visible motion and geometry, although different combinations can produce similar depth trajectories while implying different contact forces. With one sequence and no comparison against competing systems on common

data, the experiment does not establish state-of-the-art identification accuracy or uncertainty across repeated preparations. It demonstrates end-to-end fitting and measurable improvement over frozen geometry in the same recorded replay.

V. CONCLUSION

TaichiDough connects a calibrated RGB-D recording and measured two-spatula motion to floor-aware volumetric reconstruction, MLS-MPM replay, a differentiable partial-view objective, and bounded parameter optimization. On one Episode 18 sequence, the preliminary five-parameter history reduced its training objective by 61.8% over 15 accepted updates and reached a late-stage plateau. A separate fixed-configuration run improved a strict visible-geometry score over frozen initial geometry during both its fitting interval and later frames of the same recorded replay. These results support the implemented observation-conditioned fitting pipeline, but not state-of-the-art identification accuracy or unique recovery of intrinsic dough properties.

The estimated values remain conditional on the camera and floor calibration, reconstructed hidden volume, registered tool geometry, contact and friction model, constitutive parameterization, numerical discretization, and selected trajectory. Future work will add synchronized force or torque sensing, repeated dough preparations, deliberately informative robot-controlled motions, explicit tool occlusion, and evaluation on independent manipulations. Once these measurements establish predictive validity beyond a single replay, the

simulator can support synthetic-data generation and bimanual shaping policies.

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